Acid incubation study

Raw recordings to reviewed events and publication figures

Follow one page at a time. Orange numbers on each screenshot match the numbered actions beside it. Numbers restart on the next page.

01

Prepare

Identify each ABF and apply the exact recording recipe.

02

Review

Check detections; optionally edit individual events or whole clusters.

03

Analyze & style

Recompute, make the complete figure set, style it and save.

Choose your route. Keep the automatic event decisions to reproduce the saved reference. For a curated analysis, use a separate workspace copy and rerun the affected downstream stages.

The Acid reference uses calibrated pulse/level CUSUM and frozen reference families. No trained model is required.

Launcher and shared curation views are refreshed for build 40.15.59. Unchanged reference illustrations retain their capture dates. Examples show saved or explicitly labeled fixture data; this documentation update did not rerun the scientific studies.

Find the next step

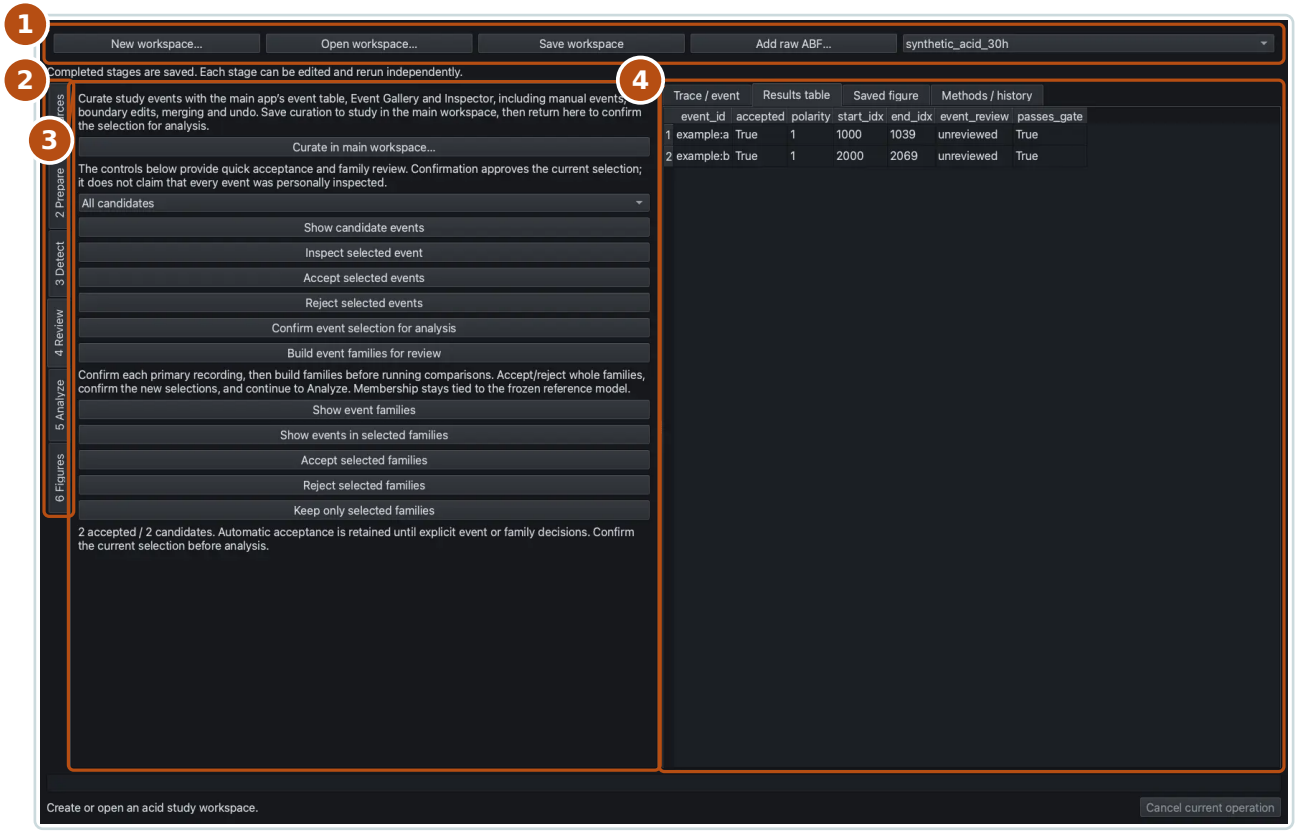
Use the page IDs in the text to find the corresponding walkthrough or recipe.

AC01	Create or reopen an incubation study	4	**AC07**	Save event edits back to the study	11
AC02	Identify the primary recordings	5	**AC08**	Confirm each current selection	12
AC03	Prepare and inspect each current trace	6	**AC09**	Review complete event families	13
AC04	Calibrate and detect candidates	7	**AC10**	Analyze confirmed events	14
AC05	Review events in the main workspace	8	**AC11**	Generate and inspect study figures	15
AC06	Inspect and correct individual events	9	**AC12**	Save styling and export a figure	16
AC06b	Review waveforms in the Event Gallery	10	**AC13**	Save and reopen the complete study	17

First, see the workspace overview on page 3. Recipes and checkpoints follow the walkthrough: exact file names, scientific settings, counts and the figure inventory. Use the PDF bookmarks to jump directly to any walkthrough or recipe.

Find your way around the workspace

Study controls and result views. Use the following close-ups for the current launcher and shared event-curation steps.



Current Acid study window · synthetic example for control orientation

1 Choose the recording

The top toolbar creates, opens and saves the workspace; the selector chooses the active recording.

2 Choose the stage

The six tabs run from Sources to Figures. If a tab is hidden, use the small arrow at the bottom of the tab strip.

3 Set and run

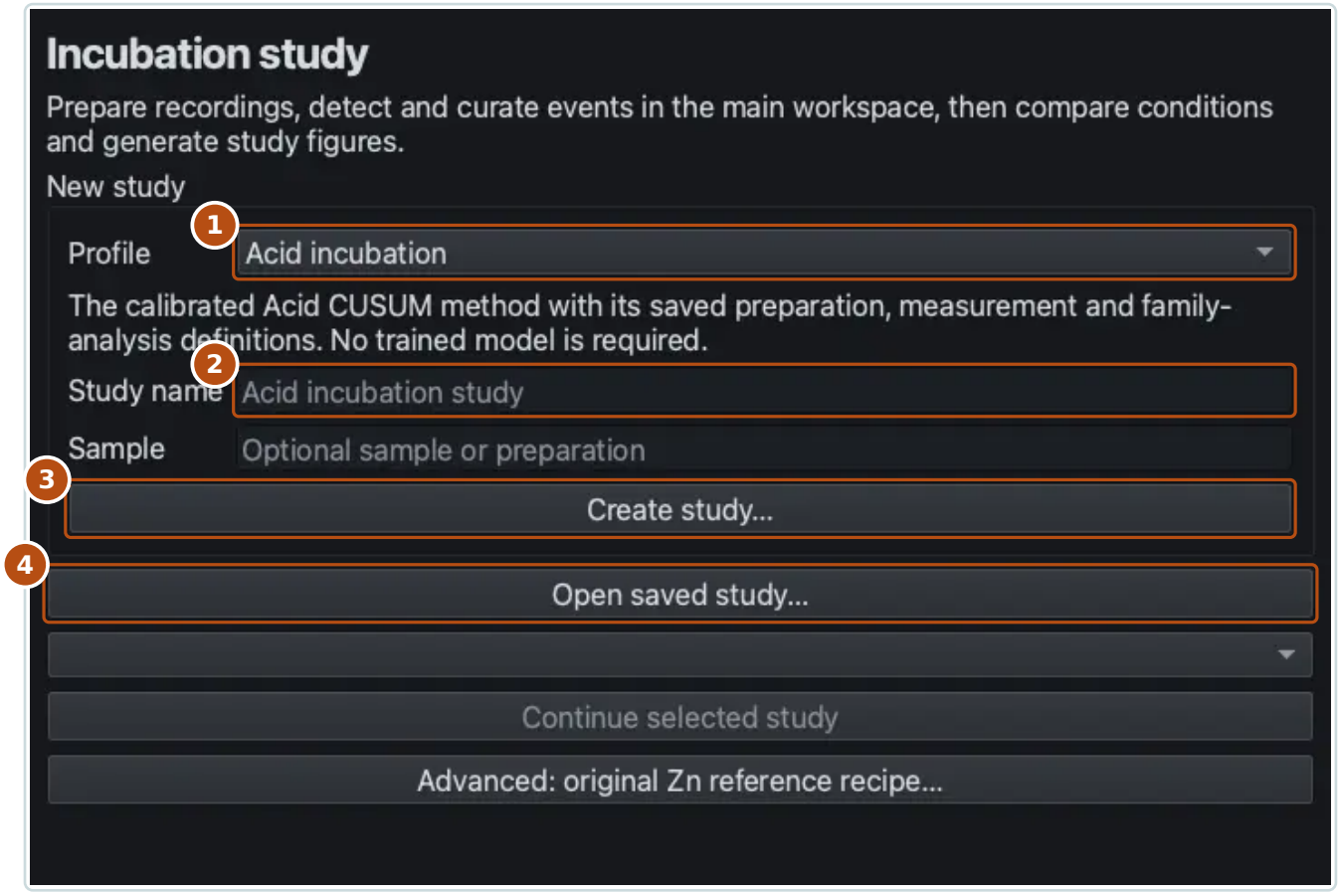
Enter values and click the stage button here. Scroll inside this panel to reach the lower controls.

4 Inspect the result

Switch among the trace, tables, figures and methods here. These tabs display the saved results.

Create or reopen an incubation study

Unified incubation workflow | NanoScope 40.15.59



Current unified study launcher · acid

DO THIS IN NANOSCOPE

1. **Choose the profile**
   Open Analysis > Incubation study... in the main app. Choose Acid incubation. Use General incubation for another sample or condition.

2. **Identify this study**
   Enter a useful study name and the sample, such as Acid-insulin. These labels do not infer or validate experimental settings.

3. **Create a new study**
   Click Create study... and choose a new empty folder. This creates the workspace; do not create it a second time in the study window.

4. **Or reopen a saved study**
   Click Open saved study... and choose the folder containing workspace.json. NanoScope restores its saved method and settings automatically.

Keep in mind

The profile supplies starting defaults. Apply every recording-specific recipe on the following pages. Saved workspaces retain their own numerical settings.

You should see

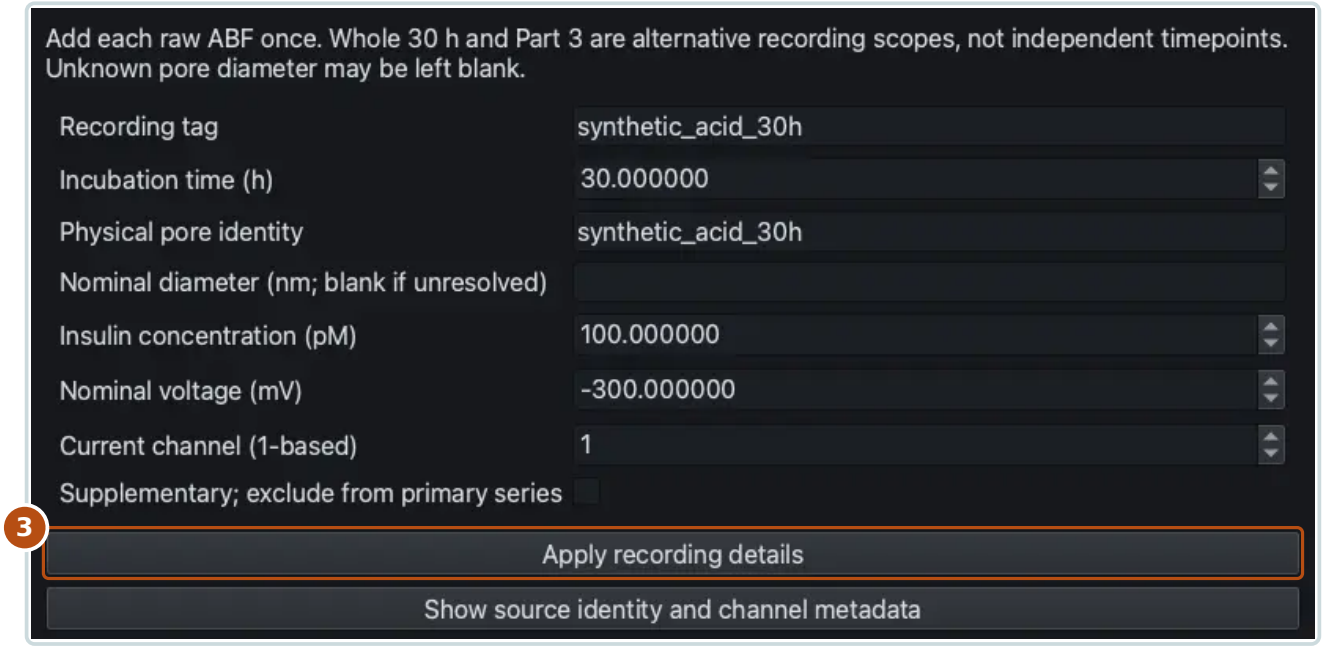
A named study opens with its staged controls. Continue selected study returns to an already open workspace.

Identify the primary recordings

Unified incubation workflow | NanoScope 40.15.59



Current Acid toolbar, cropped. Synthetic example recording.



Current build 40.15.59. Acid Sources · synthetic example; enter the experimental metadata explicitly

Values for this step
Channel: 1 (pA)
Concentration / voltage: 100 pM / -300 mV
Incubation hours: 2, 10, 18, 30

DO THIS IN NANOSCOPE

1. **Add the four primary ABFs**
   Click Add raw ABF... and select the 2, 10, 18 and whole 30-hour recordings in the input appendix. The launcher has already created your workspace folder.

2. **Choose one recording**
   Use the selector to choose the file whose metadata you are editing. Assign the exact reference tag when reproducing the saved study.

3. **Apply its identity**
   Enter the actual incubation time, distinct physical pore ID and recording conditions. Apply recording details before preparing. Leave the 30-hour diameter blank.

Keep in mind

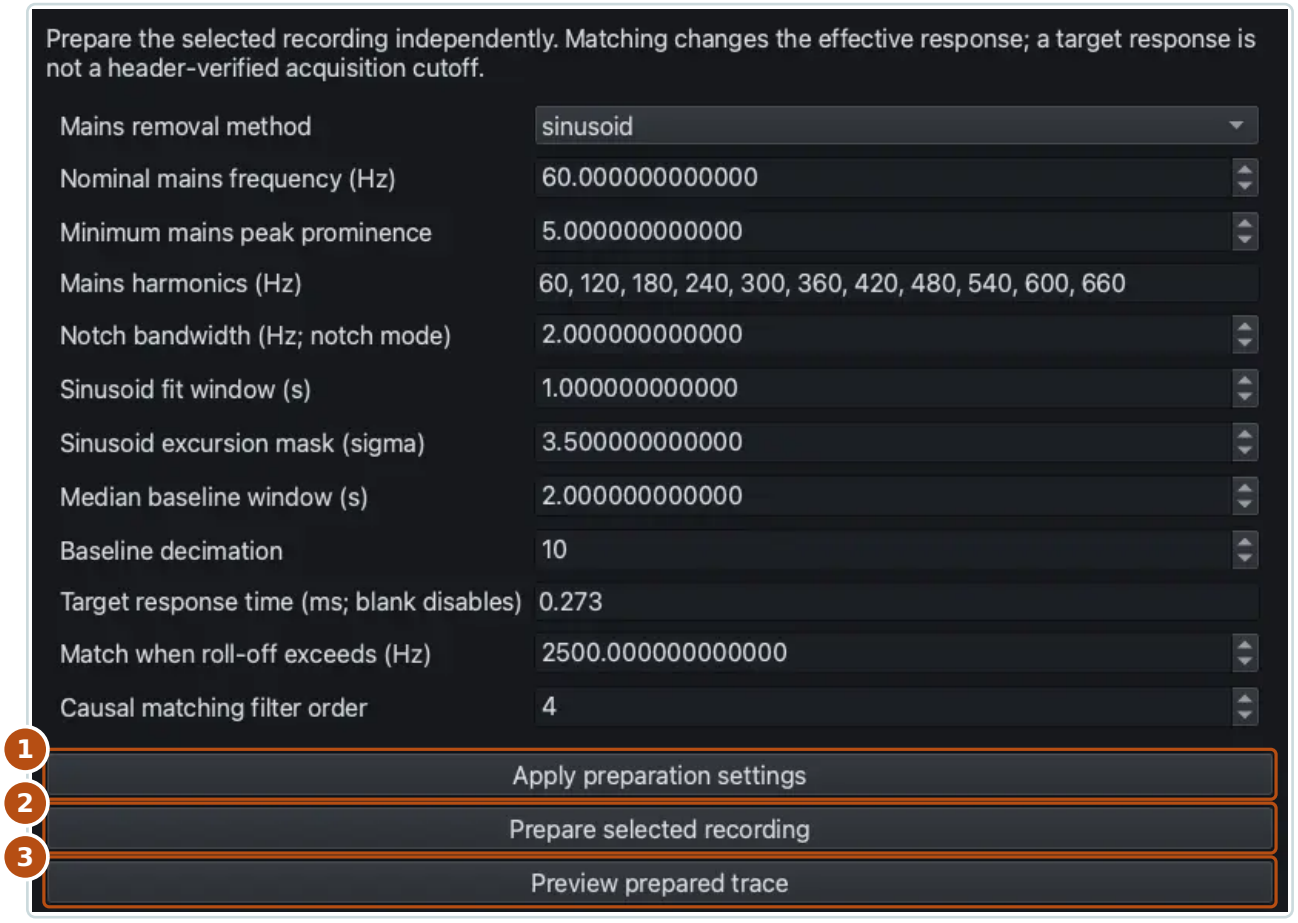
The 30-hour filename says 16 nm while the notebook labels 26 nm; this remains unresolved. Its Part 3 excerpt and baseline-only recording are supplementary checks, not independent timepoints.

You should see

Four primary recordings are present, with correct identities and source paths.

Prepare and inspect each current trace

Unified incubation workflow | NanoScope 40.15.59



Current build 40.15.59. Acid Prepare · apply settings, prepare, then inspect the trace

Values for this step
Baseline: 2 s median; decimation 10
Mains: Sinusoid; 60 Hz; prominence 5; 1 s windows; 3.5 sigma mask
Response matching: 0.273 ms, only when measured roll-off exceeds 2500 Hz; order 4

DO THIS IN NANOSCOPE

1. **Apply preparation settings**
   Use the reference recipe in Appendix PREP. If you change a setting, click Apply preparation settings before running the stage.

2. **Prepare this recording**
   Click Prepare selected recording. Allow the operation and checkpoint to finish. Repeat separately for each primary recording.

3. **Inspect the result**
   Click Preview prepared trace. Inspect the source and method metadata; use individual event inspection later for exact sample-level waveforms.

Keep in mind

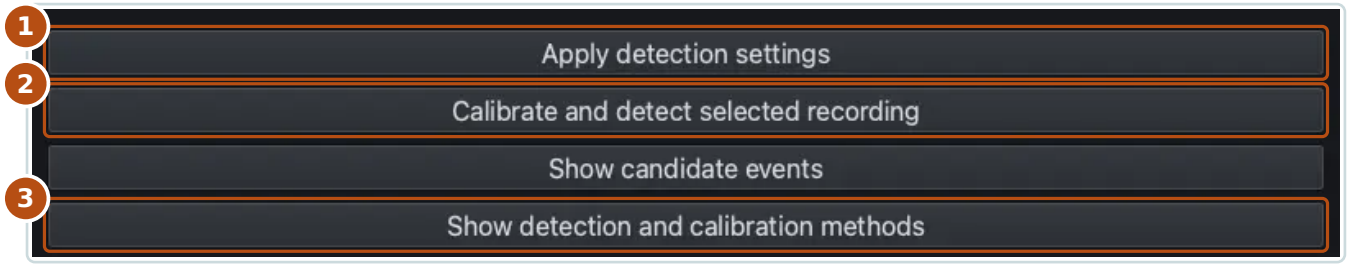
Conditional mains removal may leave the signal unchanged when no qualifying line is detected. Response matching is an analysis assumption, not proof of the original acquisition bandwidth.

You should see

The prepared trace and recorded preparation settings are available.

Calibrate and detect candidates

Unified incubation workflow | NanoScope 40.15.59



Current build 40.15.59. Acid Detect · calibration and acceptance settings are explicit · scroll continuation

Values for this step
Primary set: cusum, with matched-detection fallback enabled
Gate / reporting: Timescale peak gate, 4 sigma, local baseline; reported peak uses global baseline

DO THIS IN NANOSCOPE

1. **Apply the selected detector settings**
   Use both pulse and level scales and the saved injection/grid settings in Appendix DETECT. Apply any changes before running.

2. **Run calibration and detection**
   Click Calibrate and detect selected recording. The engine uses quiet segments and injected events to select detector settings.

3. **Inspect calibration evidence**
   Open Show detection and calibration methods. Read the calibration outcome and any fallback to matched detections.

Keep in mind

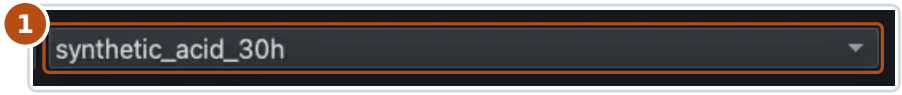
Repeat for all four recordings. Counts from the saved reference are checkpoints, not targets for threshold tuning. Empirical null rates are not a guarantee of zero false positives.

You should see

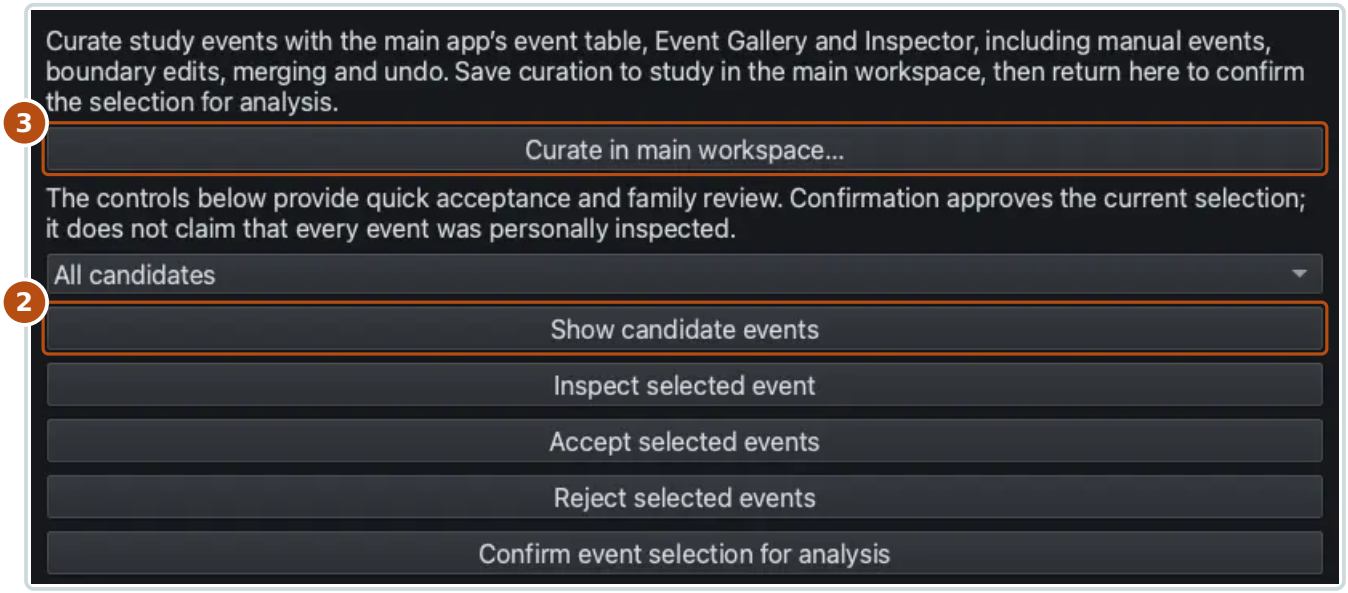
Candidate intervals and automatic selections are saved with their calibration provenance.

Review events in the main workspace

Unified incubation workflow | NanoScope 40.15.59



Current Acid recording selector. Synthetic example.



Current build 40.15.59. Acid Review · same main-app curation tools; return here to confirm selection

DO THIS IN NANOSCOPE

1. **Select the study recording**
   Choose the prepared and detected source you want to curate. Protect the automatic reference by using a complete workspace copy for an alternative selection.

2. **Inspect the candidate list**
   Show candidate events includes the automatically rejected candidates when All candidates is selected. Inspect this list before opening the main workspace.

3. **Open the main curation tools**
   On 4 Review, click Curate in main workspace.... The main app receives the attached recording and events.

Keep in mind

The main Event Table, Gallery and Inspector are shared with the rest of NanoScope. The study preview remains useful for quick review; it is not the editable boundary tool.

You should see

The main workspace shows the study bar, source identity and events.

Inspect and correct individual events

Unified incubation workflow | NanoScope 40.15.59



Current build 40.15.59. Actual Zn event · Inspector acceptance, editable boundaries and merge tools

DO THIS IN NANOSCOPE

1 Navigate by event
Use Prev and Next and check the event identity. Compare the shaded interval with the waveform.

2 Set acceptance
Use Accept or Reject to record your decision. These controls do not automatically advance to another event.

3 Correct boundaries when justified
Drag an event-mask edge and wait for remeasurement. Use Edit > Undo for supported edits. Other native editing tools can merge events or mark a missed event.

Keep in mind

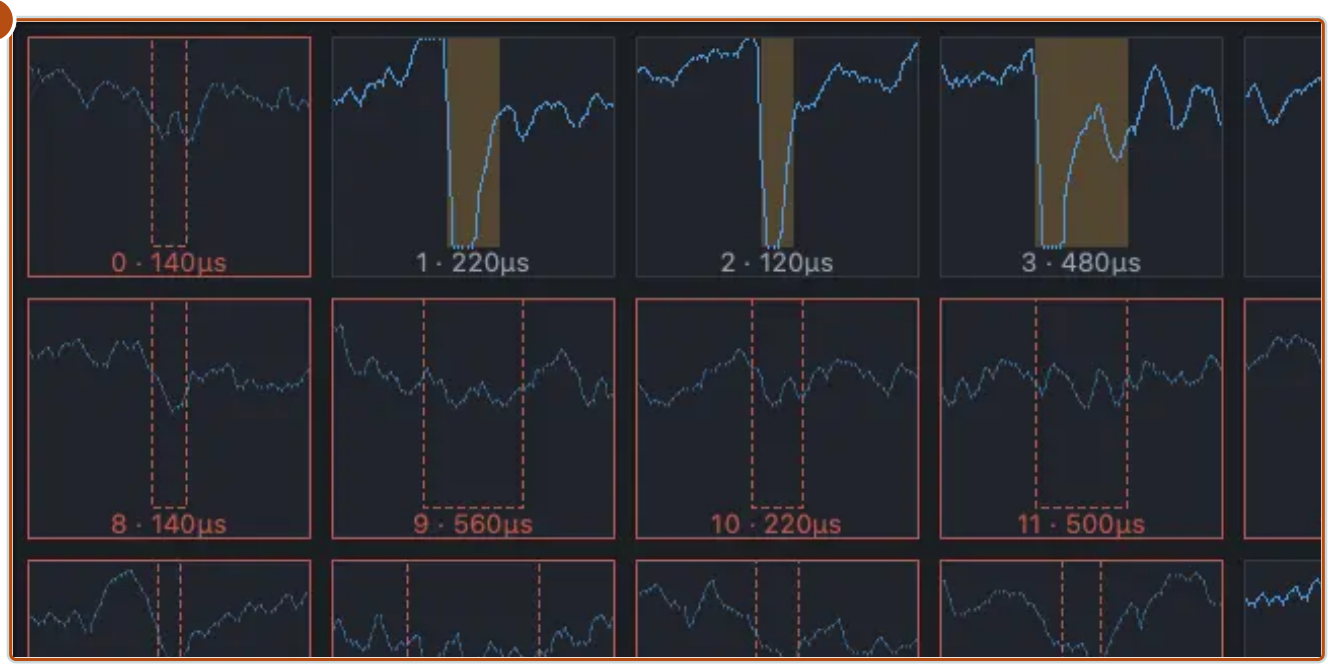
Edits must be saved to the attached study. Native selected-amplitude controls do not silently replace the acid study measurement definitions.

You should see

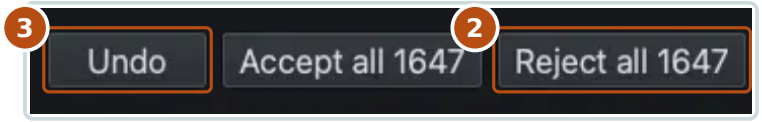
The main event table holds the intended decisions and bounds.

Review waveforms in the Event Gallery

Unified incubation workflow | NanoScope 40.15.59



Current Zn Gallery: cropped example tiles; remaining tiles are outside this crop.



Current Gallery bulk controls and Undo; read the full filter count.

DO THIS IN NANOSCOPE

1. **Compare the event tiles**
   From the main Analysis menu choose Event gallery.... Compare waveforms and open an event in the Inspector for precise boundary edits.

2. **Check the scope before a bulk decision**
   The all-events acceptance controls act on the current filter across pages. Read the displayed count. Use Accept selected or Reject selected for picked tiles instead.

3. **Undo an unintended edit**
   The Gallery Undo button uses the same undo history as the main workspace. Check the resulting event decisions after undoing.

Keep in mind

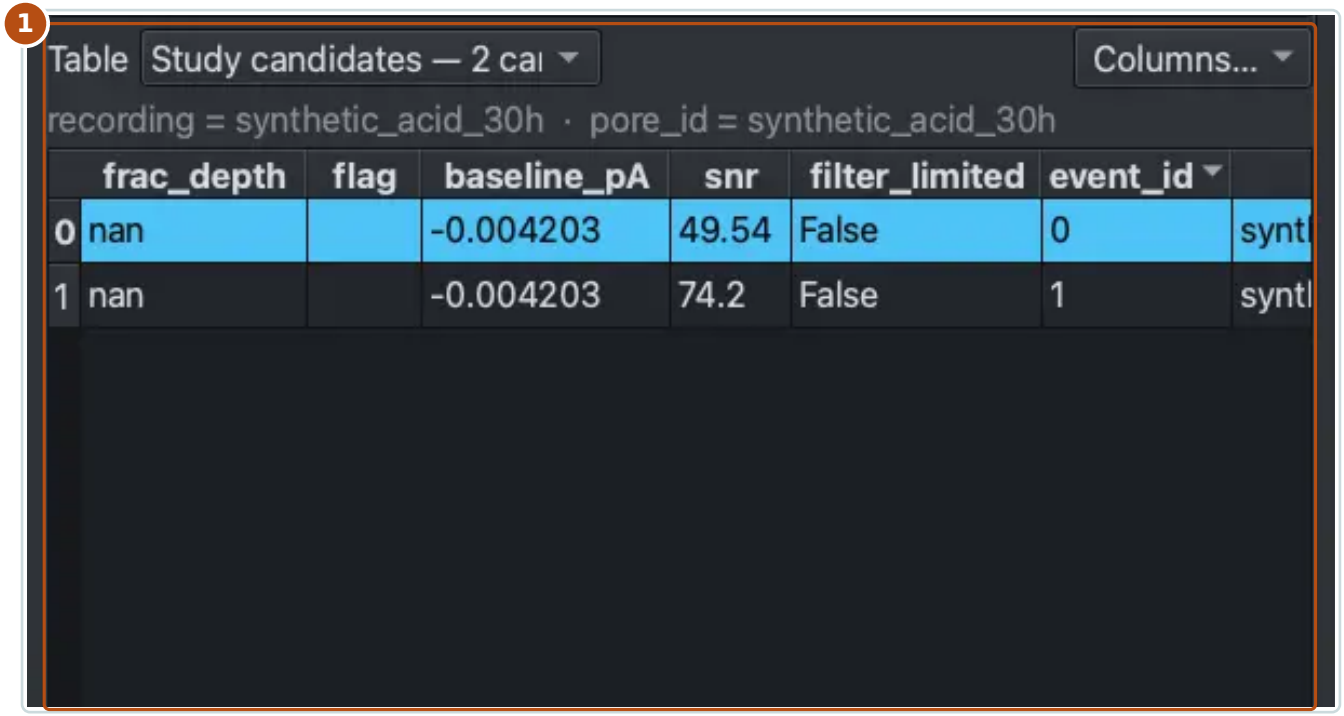
The screenshot is a cropped current-build view of saved Zn reference events. These controls are shared by all profiles. Reference reproduction retains its original acceptance decisions.

You should see

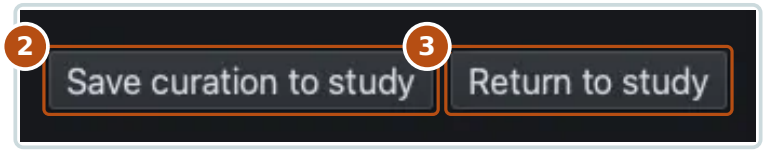
After Gallery edits, return to the main workspace and use Save curation to study before returning and confirming.

Save event edits back to the study

Unified incubation workflow | NanoScope 40.15.59



Current event table, cropped to its identity and first rows.



Current study bar: save the curation before returning.

DO THIS IN NANOSCOPE

1. **Check the active events**
   Verify that the event table belongs to the recording you opened from the study. Review acceptance and edited bounds.

2. **Commit this curation**
   Click Save curation to study and wait for completion. Acceptance-only changes preserve study measurements; edited event boundaries are remeasured from the study signals.

3. **Return to the study**
   Click Return to study. Inspect the imported decisions and confirm the selection before recomputing affected analyses.

Keep in mind

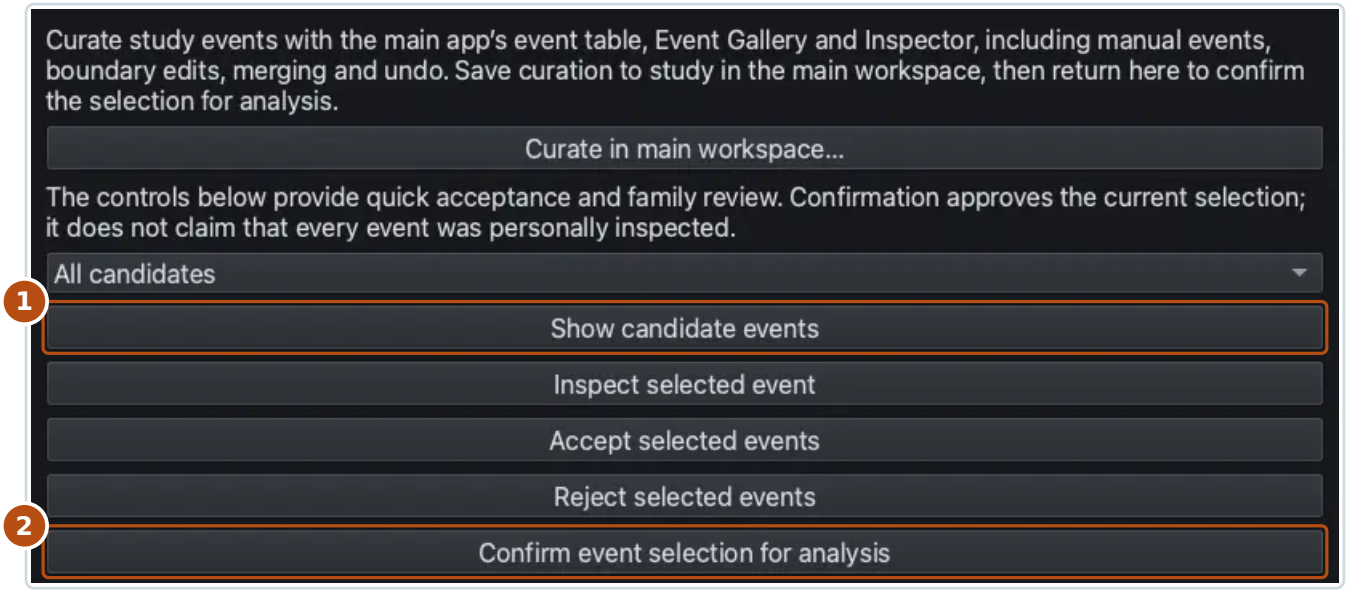
Save Project retains the main analysis session; it does not replace Save curation to study. Later event edits require another save. A failed or stale save is not a successful transfer.

You should see

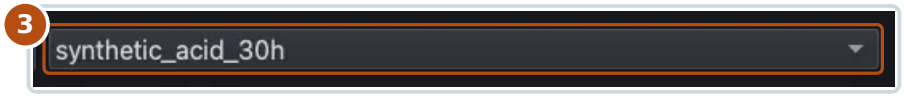
Acceptance and bounds are committed to the attached study recording. Dependent results must be recomputed after selection changes.

Confirm each current selection

Unified incubation workflow | NanoScope 40.15.59



Current build 40.15.59. Acid Review · same main-app curation tools; return here to confirm selection



Current Acid recording selector. Synthetic example.

DO THIS IN NANOSCOPE

1. **Inspect the saved candidates**
   Click Show candidate events after returning. Verify acceptance, event boundaries and identity for several edited events.

2. **Confirm the selection**
   Click Confirm event selection for analysis. Do this for every primary recording, including when keeping all automatic decisions unchanged.

3. **Choose the next recording**
   Use the selector to repeat review and confirmation for the other primary recordings.

Keep in mind

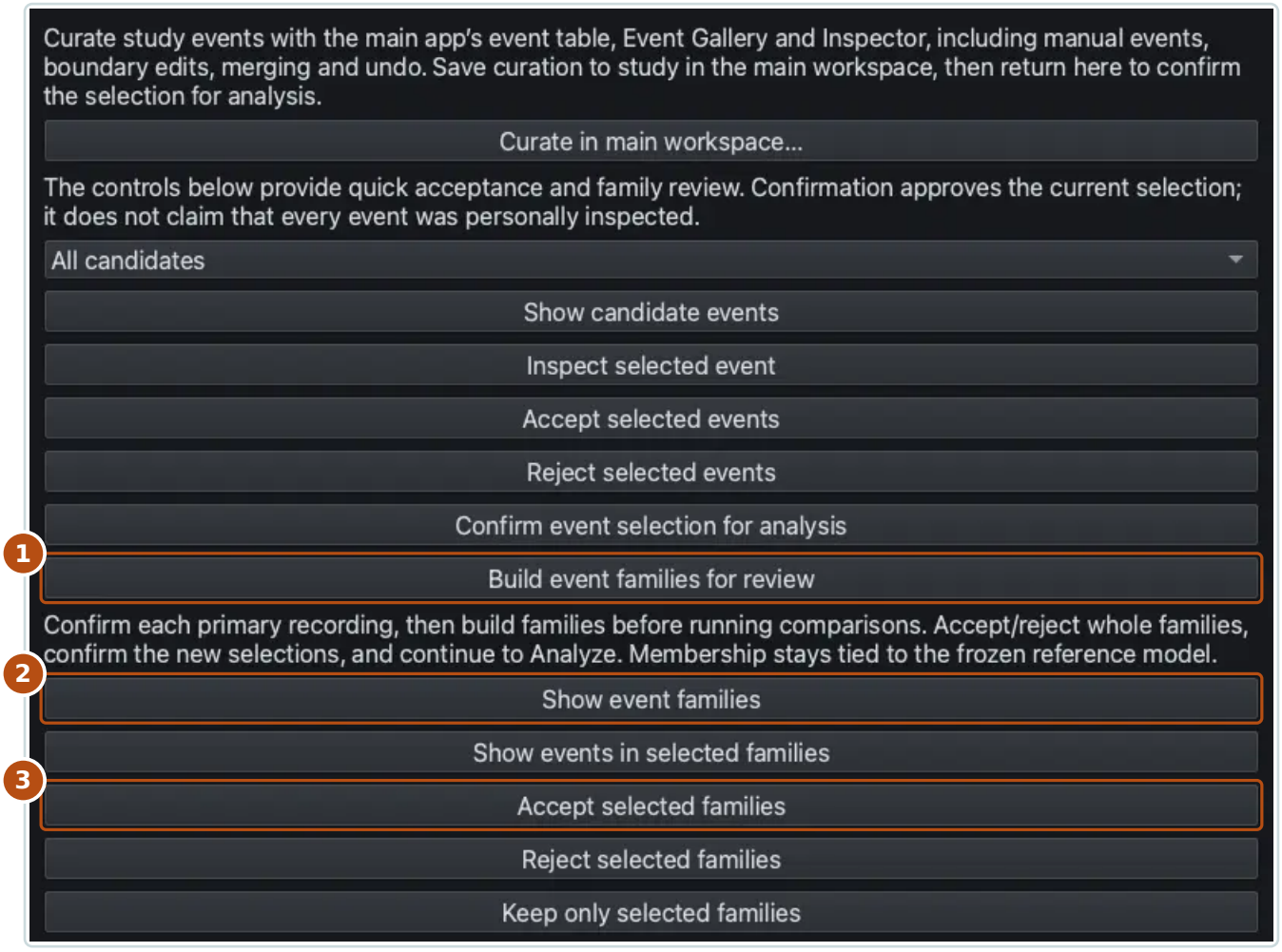
Confirmation approves the current selection; it does not claim every event was personally inspected. Any subsequent acceptance, boundary, preparation or detection change requires renewed confirmation.

You should see

Every primary recording has a current selection approval before the analysis stage.

Review complete event families

Unified incubation workflow | NanoScope 40.15.59



Current build 40.15.59. Acid Review · event selection and family decisions

DO THIS IN NANOSCOPE

1. **Build the reference families**
   After confirming all primary recordings, click Build event families for review. This saves memberships for review before the full downstream analysis.

2. **Inspect the selected family**
   Show event families, select a row, then Show events in selected families to inspect its member waveforms.

3. **Change acceptance deliberately**
   Accept or Reject selected families, or Keep only selected families. Keep-only also excludes other families and unassigned events in this recording.

Keep in mind

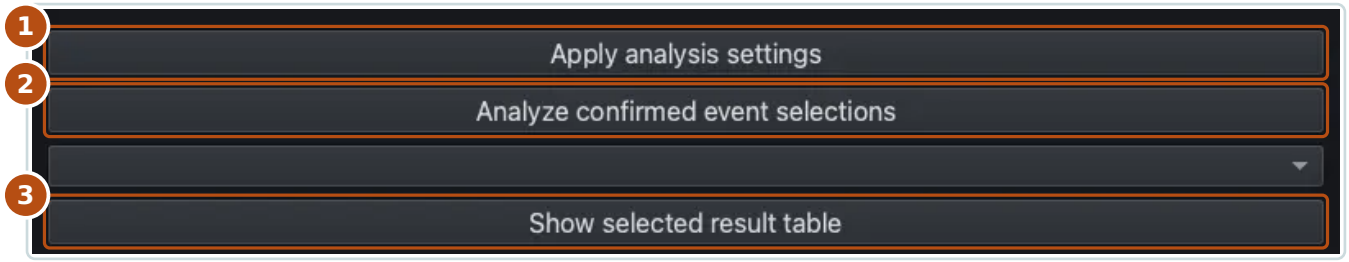
Confirm every changed recording again, then analyze. Membership remains tied to the frozen reference model, so rejecting a family does not redefine its label. Family numbers are descriptive groups, not molecular species.

You should see

Your intended individual and family decisions are saved and reconfirmed.

Analyze confirmed events

Unified incubation workflow | NanoScope 40.15.59



Current build 40.15.59. Acid Analyze · confirmed selections, validation and model settings · scroll continuation

Values for this step
Modeling minimum: 100 accepted events
Morphology: Enabled; dwell at least 0.5 ms; at least 100 events
Distribution cohort: Excludes flagged bursts; rate counts retain those primary events

DO THIS IN NANOSCOPE

1. **Apply the analysis recipe**
   Use acid_30h_whole as the reference. Keep reference refitting off after the first saved fit. Apply settings after deliberate changes.

2. **Run the analysis**
   Click Analyze confirmed event selections. Comparisons, validation and family results use the current confirmed selection.

3. **Read the result tables**
   Choose a result table and click Show selected result table. Distinguish primary, nonburst distribution and modeling cohorts.

Keep in mind

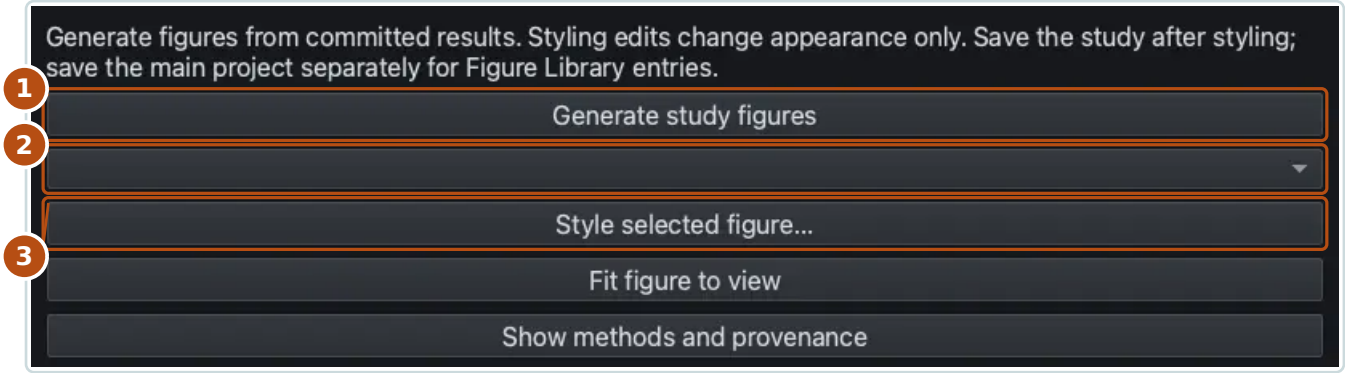
Eight time blocks and 200 block-bootstrap draws estimate within-recording uncertainty. One pore/acquisition per timepoint does not establish independent biological replication.

You should see

Saved comparison and model results are ready for figure generation.

Generate and inspect study figures

Unified incubation workflow | NanoScope 40.15.59



Current build 40.15.59. Acid Figures · generate from committed results, then choose and style a figure

DO THIS IN NANOSCOPE

1. **Generate from final results**
   Click Generate study figures after completing all scientific changes. Old files may remain in prior runs; use the current saved figure selector.

2. **Select a figure**
   Choose an entry in the saved-figure selector and inspect it in Saved figure. Read its method and cohort qualifications.

3. **Open the style editor**
   Click Style selected figure... to adjust appearance without changing the accepted events or numerical results.

Keep in mind

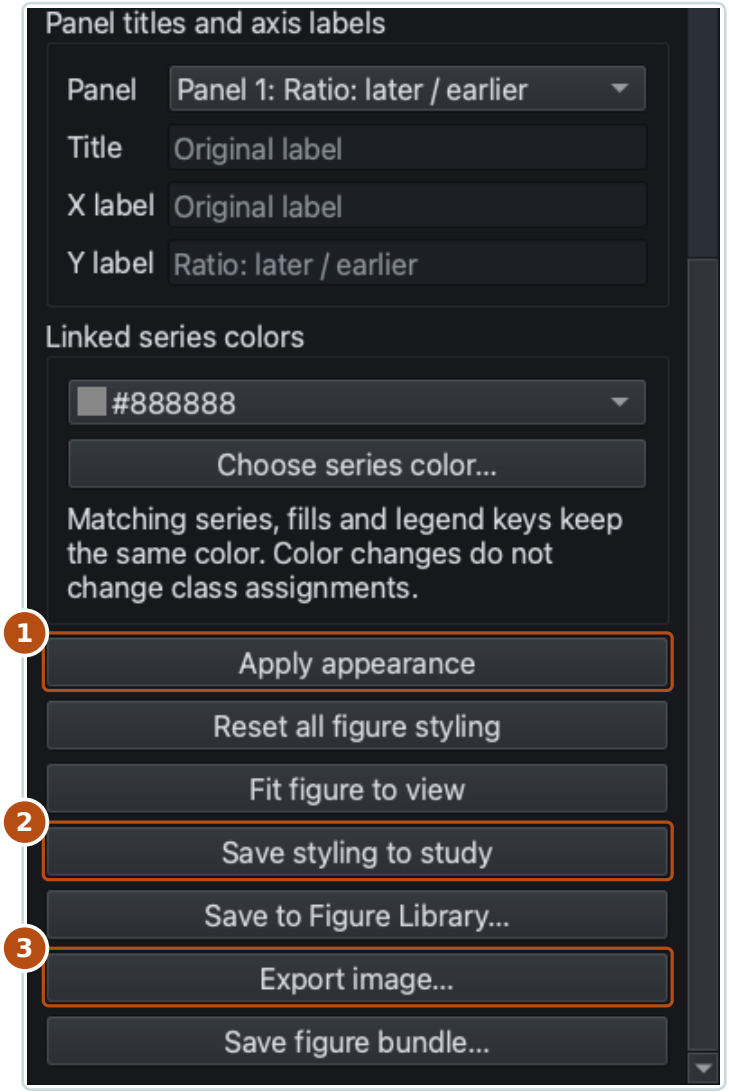
The original automatic baseline contains 127 scientific figure entries. A curated or insufficient-data analysis can produce a different eligible set. Do not force a count by changing scientific thresholds.

You should see

Current figures are available with their numerical recipes and provenance.

Save styling and export a figure

Unified incubation workflow | NanoScope 40.15.59



Reference capture, 21 Sep 2026. NanoScope reference controls and saved results.

DO THIS IN NANOSCOPE

1. **Apply appearance**
   Set the desired typography, panel labels and linked series colors, then Apply appearance to preview.

2. **Save to the study**
   Click Save styling to study so the appearance survives reopening and regeneration.

3. **Export the current figure**
   Use Export image... or Save figure bundle.... Save to Figure Library... is available for publication composition.

Keep in mind

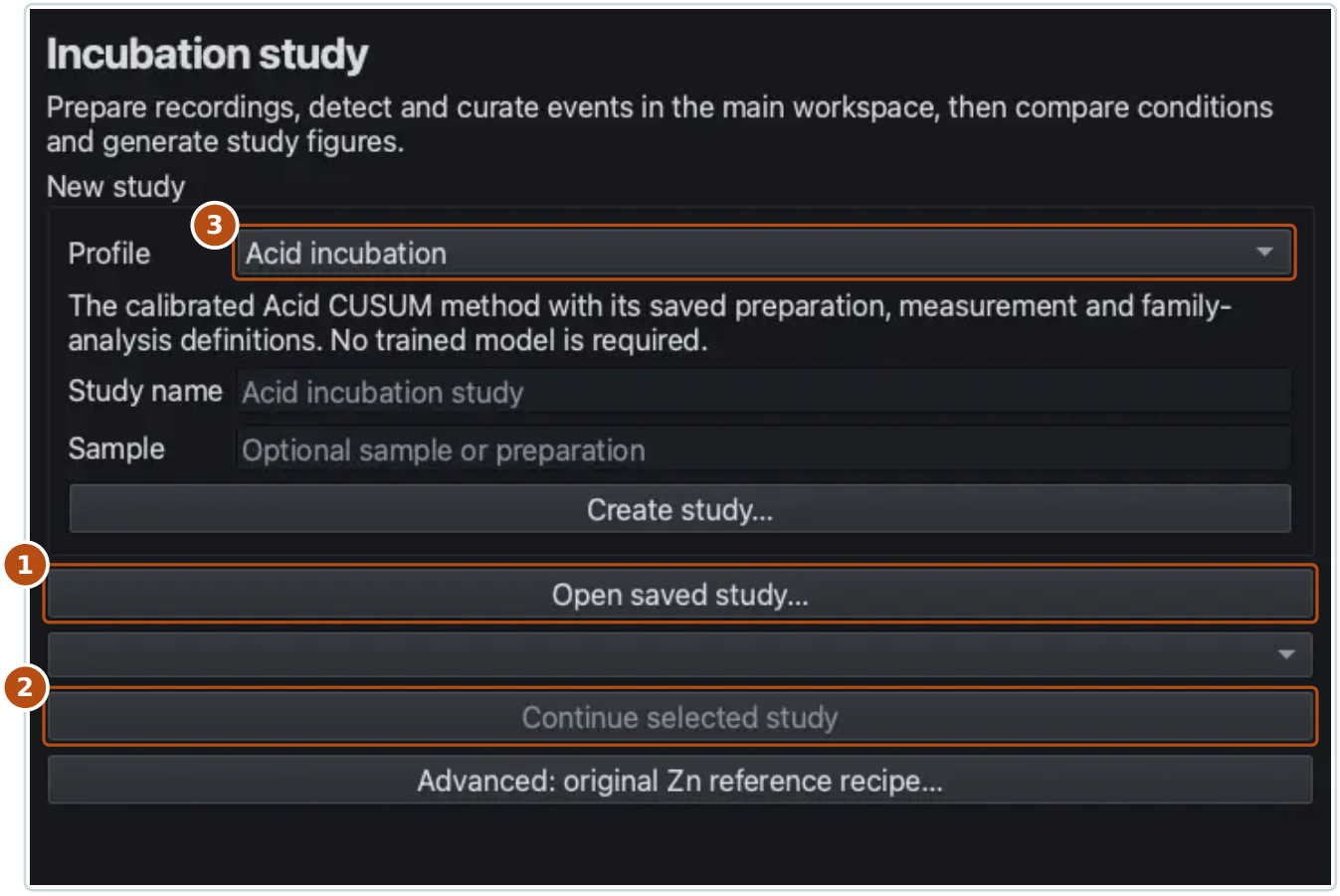
Figure Library entries belong to the main NanoScope project: save that project separately. General styling is preserved, but custom event captions must still match the event identity and measurements.

You should see

The figure appearance is stored, and the exported file matches the preview.

Save and reopen the complete study

Unified incubation workflow | NanoScope 40.15.59



Current unified study launcher · acid

DO THIS IN NANOSCOPE

1. **Reopen the saved folder**
   In the study, finish any operation and click Save workspace. Close it, then open Analysis > Incubation study... > Open saved study.... Select the whole workspace folder.

2. **Continue an open study**
   If the study is already open, select it in Open study workspaces and click Continue selected study.

3. **Keep the saved method**
   The Profile control is for new studies. Opening a saved study selects the proper scientific method without applying a different profile.

Keep in mind

Keep workspace.json, arrays, runs and supporting files together. Preserve original ABFs and models for reruns. Use one editing session per folder. Save the main project separately for main curation and Figure Library entries.

You should see

Check restored event decisions, committed settings, comparison tables, fits and styled figures without rerunning completed stages.

INPUTS / Primary inputs and metadata

Original sources are in /Users/abrahamcampos/Desktop/OnlyAcidCondition/. The saved automatic reference is /Users/abrahamcampos/Documents/Codex/2026-09-12/le/outputs/acid-incubation-20260928/manual-study. Select local copies on another computer.

All primary acquisitions use current channel 1, recording concentration 100 pM and nominal voltage -300 mV. Give each distinct physical pore/acquisition a separate identity. Do not infer that the recording concentration equals the incubation stock concentration.

This manual documents data analysis. Incubation temperature, stock preparation, metal stoichiometry, pH and mixing conditions must come from the laboratory record.

Tag / hours	Exact raw ABF filename	Nominal pore diameter
acid_2h / 2 h	Insulin 100 pm_2h_18 nm_50 PEG__0005_Process_303 s__used for event rate_13 pA for base line_Lab view corrected.abf	18 nm
acid_10h / 10 h	10h_100 pM_50 percent PEG_20 nm_0000_0003_Conductive_-300 mV_Process_380 s_used for event rate_11 pA for base line_Lab view corrected.abf	20 nm
acid_18h / 18 h	18h_100 pM_50 percent PEG_18 nm_0000_0002_-Conductive_-300 mV_Process_part_01_used for event rate_356 s_10 pA for base line_Lab view corrected.abf	18 nm
acid_30h_whole / 30 h	30 h_100 pM_ 16 nm_50 percent_35k_PEG_0000_whole time point.abf	Unresolved; leave blank

PREP / Preparation reference settings

Controls are editable. These values reproduce the saved model-free reference, not universal defaults for every future sample.

Control / checkpoint	Value
Mains removal method	sinusoid
Nominal mains frequency (Hz)	60.0
Minimum mains peak prominence	5.0
Mains harmonics (Hz)	60, 120, 180, 240, 300, 360, 420, 480, 540, 600, 660
Notch bandwidth (Hz; notch mode)	2.0
Sinusoid fit window (s)	1.0
Sinusoid excursion mask (sigma)	3.5
Median baseline window (s)	2.0
Baseline decimation	10
Target response time (ms; blank disables)	0.273
Match when roll-off exceeds (Hz)	2500.0
Causal matching filter order	4

DETECT / Detection and calibration reference settings

Controls are editable. These values reproduce the saved model-free reference, not universal defaults for every future sample.

Control / checkpoint	Value
Also detect upward / resistive events	On
Combine pulse and level scales	On
Accepted false-alarm budget (/s)	0.02
Raw false-alarm cap (/s)	0.5
Quiet calibration exposure (s)	150.0
Pulse baseline block (ms)	2.0
Level baseline block (ms)	50.0
Pulse drift grid	3.0, 4.0, 5.0, 6.0, 8.0
Pulse evidence grid	5.0, 10.0, 20.0, 30.0
Level drift grid	2.0, 3.0, 4.0, 5.0, 6.0
Level evidence grid	20.0, 45.0, 70.0, 100.0, 150.0
Injection depths (pA)	4.0, 6.0, 8.0, 12.0, 20.0
Injection dwells (ms)	0.1, 0.2, 0.5, 1.0, 5.0
Injections per depth/dwell cell	40

Control / checkpoint	Value
Use matched detections if calibration fails	On
Primary detection set	cusum
Acceptance depth statistic	peak
Acceptance noise rule	timescale
Acceptance significance (sigma)	4.0
Acceptance baseline	local
Reported amplitude	peak
Reported amplitude baseline	global
Minimum event duration (ms)	0.05
Merge gap (ms)	0.05
Cross-check entry threshold (sigma)	4.0
Cross-check exit threshold (sigma)	1.5
Local context (ms)	2.0
Maximum local context (ms)	20.0

ANALYZE / Analysis reference settings

Controls are editable. These values reproduce the saved model-free reference, not universal defaults for every future sample.

Control / checkpoint	Value
Family reference recording	acid_30h_whole
Refit the reference family model	Off
Exclude flagged bursts from fits	On
Burst maximum onset gap (ms)	20.0
Minimum events in a burst	5
Minimum accepted events for modeling	100
Validation time blocks	8
Block bootstrap draws	200
Mixture component grid	1, 2, 3
Minimum mixture weight	0.05
Minimum mixture separation	1.0
Required block presence fraction	0.75
Required BIC improvement	10.0
Required held-out likelihood gain	0.0

Control / checkpoint	Value
Minimum label probability	0.8
Analyze long-event morphology	On
Morphology minimum dwell (ms)	0.5
Morphology minimum events	100
Resampled waveform points	48
Density cluster-size fractions	0.03, 0.06, 0.1
Density minimum-sample grid	10, 25, 50
Density cluster-size floor	15
Figure resolution (dpi)	180

CHECK / Saved automatic reference checkpoints

These are previously validated reference results, not a new analysis run for this manual. A curated selection intentionally changes the counts. The 2, 10 and 18-hour primary sets require matched-detector agreement after calibration fallback. The 30-hour primary set retains accepted CUSUM events.

The 21 events excluded from the 30-hour distribution plots belong to two bursts; they remain in the 849-event rate and family-count cohorts. Do not compare counts without stating the cohort.

The six descriptive reference families are transferred from the whole 30-hour recording. The original independent screening found no robust discrete subpopulation at 10, 18 or 30 hours; 2 hours had insufficient events for the 100-event modeling minimum.

Time	Candidates	Gate passing	Primary events	Nonburst distribution
2 h	245	40	22	22
10 h	291	125	110	110
18 h	445	198	142	142
30 h	1850	849	849	828

LIMITS / Interpretation, saving and future studies

Physical pore and recording differences are confounded with incubation time. Fractional current normalization does not remove this limitation. Bootstrap intervals describe within-recording uncertainty, not independent pore or biological replication.

Admission uses a local-baseline, duration-dependent peak threshold. The reported main amplitude is global-baseline peak current change divided by the time-local absolute open-pore current. These are different calculations.

In the validated reference, no qualifying mains line was detected, so the mains-removal step was skipped. The 2, 10 and 18-hour recordings receive conditional Bessel matching; the full 30-hour recording receives no additional matching filter. Its header declares a 1 kHz signal-conditioner stage, not a complete acquisition transfer function.

The 0.5 ms morphology threshold does not guarantee physical resolution. Under the chosen 3-tau convention, events must exceed 0.819 ms to be classed as resolved.

Keep automatic gate decisions separate from selected acceptance. After native edits use Save curation to study, Return to study and Confirm event selection for analysis; then rerun Analyze and Figures. Save Project and Save workspace serve different purposes.

General, Zn, Cu and Acid share the launcher and main curation tools. For a future sample choose General incubation and explicitly document its actual experimental details and suitable scientific parameters. Do not reuse a calibrated acid method solely because the experiment has incubation timepoints.